

From classical statistics to statistical learning: a probabilistic overview

I — Statistical models a la Fisher

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The starting point: observed data

Every classical statistical analysis begins with a finite collection of data:

$$x_1, x_2, \dots, x_n.$$

which we assume to be real-valued for simplicity.

The probabilistic interpretation of these data is that they are a realization of a random sample:

$$\mathbf{x} = (x_1, x_2, \dots, x_n) \rightsquigarrow \mathbf{X} = (X_1, X_2, \dots, X_n), \quad X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_X,$$

where $P_X = X_{\#}P$ is the distribution on $\mathbb{R}$ induced by a random variable $X : (\Omega, \mathcal{F}, P) \rightarrow \mathbb{R}$.

Central idea

The observed data belong to the *sample level*, whereas the common distribution P_X , generally unknown, belongs to the *population level*.

Essential distinction

observed data $\{x_i\}$ $\neq$ random variables $\{X_i\}$ $\neq$ population distribution P_X .

The unknown population distribution

The statistical problem

The distribution P_X is not observed directly: what we observe is a single realization

$$x_1, \dots, x_n$$

of the random sample. Repeating the sampling experiment would generally produce a different realization, which is the source of its random character.

Thus, statistical inference, an essentially **inductive** method, is built around a fundamental asymmetry:

unknown population law P_X

→

observed data $x_1, \dots, x_n$.

The population law generates the data (we also say that the data has been drawn from the law), whereas inference proceeds in the opposite direction.

Before introducing a model

At this stage no particular form for P_X has been assumed: the population distribution may, in principle, be arbitrary.

Fisher's move: a statistical model

Fisher's fundamental move is to assume that P_X belongs to a parametric family of distributions:

$$P_X \in \mathcal{P} = \{P_\theta : \theta \in \Theta\}, \quad \Theta \subset \mathbb{R}^q.$$

Equivalently, the random sample is modeled as

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_\theta, \quad \theta \in \Theta \text{ unknown.}$$

The key reduction

The problem of determining an arbitrary unknown distribution P_X is replaced by the problem of determining the unknown parameter $\theta \in \Theta$.

The statistical model

The family

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}$$

represents the set of admissible population distributions. It is assumed that the model is **identifiable** in the sense that the map $\theta \mapsto P_\theta$ is injective.

Once a statistical model has been specified, the data are used to select a value of θ .

Estimator

An estimator of θ is a statistic

$$\hat{\theta} = h(\mathbf{X}) : \Omega \rightarrow \Theta,$$

where h does not depend on the unknown parameter θ .

For the observed realization $\mathbf{x} = (x_1, \dots, x_n)$, we obtain the estimate

$$\hat{\theta}(\mathbf{x}) = h(\mathbf{x}) \in \Theta,$$

and hence the fitted distribution $P_{\hat{\theta}(\mathbf{x})}$.

From model to fitted model

model $\{P_\theta : \theta \in \Theta\}$ + data $\mathbf{x}$ $\longrightarrow$ estimate $\hat{\theta}(\mathbf{x})$ $\longrightarrow$ fitted model $P_{\hat{\theta}(\mathbf{x})}$.

How should an estimator be evaluated?

Once an estimator $\hat{\theta} = h(\mathbf{X})$ has been proposed, two complementary questions naturally arise.

1. Performance

How well does $\hat{\theta}$ perform when compared with competing estimators? Typically, the comparison is made within a prescribed family of candidates, keeping in mind that the best estimator in that family need not be optimal in a broader sense.

2. Uncertainty

Even after an estimator has been selected, the point estimate $\hat{\theta}(\mathbf{x})$ will almost never coincide exactly with the true parameter θ . One must therefore quantify the variability of the random estimator around the true value.

Two complementary tasks

compare estimators

&

quantify uncertainty for a chosen estimator

Mean squared error

A natural way to evaluate an estimator $\hat{\theta}$ is to measure how far it typically lies from the true parameter θ .

Mean squared error

The mean squared error of $\hat{\theta}$ is

$$\text{mse}(\hat{\theta}) = \mathbb{E}(\|\hat{\theta} - \theta\|^2).$$

Thus, within a given family of competing estimators, smaller MSE means better performance.

Bias-variance trade-off

$$\text{mse}(\hat{\theta}) = \text{tr } \mathbb{C}(\hat{\theta}) + \|\text{bias}(\hat{\theta})\|^2,$$

where $\mathbb{C}(\hat{\theta})$ is the covariance matrix of $\hat{\theta}$ and

$$\text{bias}(\hat{\theta}) = \mathbb{E}(\hat{\theta}) - \theta$$

is its bias.

Asymptotic normality

Let

$$\hat{\theta}_n \in \mathbb{R}^q, \quad \theta \in \Theta \subset \mathbb{R}^q,$$

be an estimator based on a random sample of size n .

Asymptotic normality

The estimator $\hat{\theta}_n$ is asymptotically normal if, for each $\theta \in \Theta$,

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}_q(0, \Sigma_\theta),$$

where Σ_θ is a positive definite $q \times q$ asymptotic covariance matrix.

Asymptotic normality implies consistency

$$\sqrt{n}(\hat{\theta}_n - \theta) = O_P(1) \implies \hat{\theta}_n - \theta = O_P(n^{-1/2}) \xrightarrow{P} 0$$

Moreover, it identifies the natural $n^{-1/2}$ scale of the estimation error.

We have already seen that asymptotic normality implies consistency:

$$\hat{\theta}_n \xrightarrow{P} \theta.$$

Adding control of dispersion to consistency

If the covariance matrices remain uniformly bounded, then consistency implies

$$\mathbb{E}_\theta(\hat{\theta}_n) \longrightarrow \theta.$$

Thus, $\hat{\theta}_n$ is asymptotically unbiased (that is, $\text{bias}_\theta(\hat{\theta}_n) \longrightarrow 0$).

This naturally leads to the question of how small the covariance matrix of an **unbiased** estimator can be.

The next question

Among unbiased estimators, how small can the covariance matrix be?

Fisher information

If $X_1, \dots, X_n$ is an iid sample and $\mathbf{x} = (x_1, \dots, x_n)$ is an observed realization of the random vector $\mathbf{X} = (X_1, \dots, X_n)$ with $dP_{X_j} = \psi_\theta(x) dx$, $\theta \in \Theta \subset \mathbb{R}^q$, then

$$L(\mathbf{x}; \theta) = \prod_{i=1}^n \psi_\theta(x_i),$$

is the likelihood function (the joint density of the sample) and

$$\ell(\mathbf{x}; \theta) = \log L(\mathbf{x}; \theta)$$

is the associated log-likelihood function.

The corresponding *score vector* is

$$s(\mathbf{X}; \theta) = (\nabla_\theta \ell)(\mathbf{X}; \theta).$$

Fisher information matrix

The Fisher information carried by the sample about the parameter θ is the covariance matrix of the score:

$$\mathcal{F}_\mathbf{X}(\theta) = \mathbb{C}_\theta(s(\mathbf{X}; \theta)) = -\mathbb{E}_\theta((\nabla_\theta s)(\mathbf{X}; \theta)) = -\mathbb{E}_\theta((\nabla_{\theta\theta}^2 \ell)(\mathbf{X}; \theta)).$$

Thus, $\mathcal{F}_\mathbf{X}(\theta)$ is a $q \times q$ positive semidefinite matrix determined by the statistical model and the sample, not by any particular estimator. Usually, it is assumed that $\mathcal{F}_\mathbf{X}(\theta)$ is positive definite (and hence invertible).

Cramér–Rao inequality

Under suitable regularity assumptions, any unbiased estimator $\hat{\theta}$ of $\theta \in \Theta \subset \mathbb{R}^q$ satisfies

$$\mathbb{C}_{\theta}(\hat{\theta}) \succeq \mathcal{F}_{\mathbf{X}}(\theta)^{-1},$$

where $\mathcal{F}_{\mathbf{X}}(\theta)$ is the Fisher information matrix carried by the sample.

Thus, the statistical model itself imposes a lower bound on the dispersion of any unbiased estimator.

Efficiency

An unbiased estimator attaining the bound,

$$\mathbb{C}_{\theta}(\hat{\theta}) = \mathcal{F}_{\mathbf{X}}(\theta)^{-1},$$

is called efficient.

Maximum likelihood: Fisher's canonical estimator

Given a statistical model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, and observed data $\mathbf{x} = (x_1, \dots, x_n)$, one obtains the likelihood function $L(\mathbf{x}; \theta)$ as above.

Maximum likelihood estimator

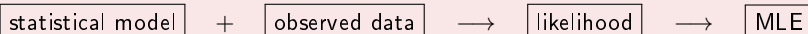
The maximum likelihood estimator selects the parameter value that makes the observed data most likely:

$$\hat{\theta}_{\text{ML}}(\mathbf{x}) = \arg \max_{\theta \in \Theta} L(\mathbf{x}; \theta),$$

whenever the maximum is attained.

Thus, maximum likelihood fits the statistical model to the observed data by selecting the parameter value $\hat{\theta}_{\text{ML}}(\mathbf{x})$.

A canonical construction



Asymptotic normality of the MLE

Consider an iid sample $X_j \sim P_\theta$ from a statistical model, and let $\hat{\theta}_{\text{ML},n}$ be a consistent MLE of $\theta \in \Theta \subset \mathbb{R}^q$.

Asymptotic normality

Under suitable regularity assumptions,

$$\sqrt{n}(\hat{\theta}_{\text{ML},n} - \theta) \xrightarrow{d} \mathcal{N}_q(0, \mathcal{F}^{(1)}(\theta)^{-1}),$$

where $\mathcal{F}^{(1)}(\theta) = \mathcal{F}^{(n)}(\theta)/n$ is the Fisher information matrix for one observation and $\mathcal{F}^{(n)}(\theta) = \mathcal{F}_X(\theta)$ as above. Equivalently,

$$\hat{\theta}_{\text{ML},n} \approx_{n \rightarrow +\infty} \mathcal{N}_q(\theta, \mathcal{F}^{(n)}(\theta)^{-1}).$$

This leads to the construction of asymptotic confidence intervals for $\theta \in \mathbb{R}$:

$$\theta \in \left[\hat{\theta}_{\text{ML},n} \mp \frac{z_{1-\delta/2}}{\sqrt{\mathcal{F}^{(n)}(\hat{\theta}_{\text{ML},n)}}} \right] \quad \text{with probability } 1 - \delta.$$

The remarkable feature

The asymptotic covariance of the MLE is determined entirely by the Fisher information of the statistical model.

Recall that the Cramér–Rao inequality gives, for an unbiased estimator $\tilde{\theta}_n$,

$$\mathbb{C}_\theta(\tilde{\theta}_n) \succeq \mathcal{F}^{(n)}(\theta)^{-1}.$$

On the other hand, for the maximum likelihood estimator,

$$\hat{\theta}_{\text{ML},n} \approx_{n \rightarrow +\infty} \mathcal{N}_q\left(\theta, \mathcal{F}^{(n)}(\theta)^{-1}\right), \quad \mathcal{F}^{(n)}(\theta) = n\mathcal{F}^{(1)}(\theta).$$

Asymptotic efficiency

Under suitable regularity assumptions, the MLE asymptotically attains the C-R bound:

$$\mathbb{C}_\theta(\hat{\theta}_{\text{ML},n}) \approx_{n \rightarrow +\infty} \mathcal{F}^{(n)}(\theta)^{-1}.$$

Keep in mind that in general $\hat{\theta}_{\text{ML},n}$ fails to be unbiased!

AIC: from fitting to model selection

For fixed observed data $\mathbf{x}$, maximum likelihood associates with each statistical model a fitted distribution $P_{\hat{\theta}_{\text{ML}}(\mathbf{x})}$.

Akaike information criterion

If the model has q free parameters, its AIC is

$$\text{AIC} = \underbrace{-2 \ell(\mathbf{x}; \hat{\theta}_{\text{ML}}(\mathbf{x}))}_{\text{fit}} + \underbrace{2q}_{\text{complexity penalty}}.$$

Thus, AIC assigns a numerical score to a model after it has been fitted to the observed data.

Model selection

Among competing models fitted to the same data,

smaller AIC is preferred.

AIC is primarily a relative model-selection criterion. Here, however, we will use it to ask a different question:
through which features of the observed data does the fitted likelihood depend on the sample?

Example: the exponential model

Let

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\lambda), \quad f_\lambda(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad \lambda > 0.$$

One has

$$L(\mathbf{x}; \lambda) = \lambda^n e^{-\lambda \sum_{i=1}^n x_i} \implies \ell(\mathbf{x}; \lambda) = n \log \lambda - n\lambda \bar{x} \implies s(\mathbf{x}; \lambda) = \frac{n}{\lambda} - n\bar{x},$$

where $\bar{x}$ is a realization of the sample mean $\bar{X}$. Thus,

$$\hat{\lambda}_{\text{ML}}(\mathbf{x}) = \frac{1}{\bar{x}},$$

Fisher information

$$\frac{d}{d\lambda} s(\mathbf{x}; \lambda) = -\frac{n}{\lambda^2} \implies \mathcal{F}^{(n)}(\lambda) = \frac{n}{\lambda^2}.$$

Exponential model: large-sample behavior

For

$$\hat{\lambda}_{\text{ML}} = \frac{1}{\bar{X}},$$

the finite-sample expectation is

$$\mathbb{E}_{\lambda}(\hat{\lambda}_{\text{ML}}) = \frac{n}{n-1}\lambda, \quad n > 1.$$

Thus, the MLE is biased, but

$$\text{bias}_{\lambda}(\hat{\lambda}_{\text{ML}}) = \frac{\lambda}{n-1} \rightarrow 0. \quad [\text{asymptotically unbiased}]$$

Moreover,

$$\hat{\lambda}_{\text{ML}} \approx \mathcal{N}\left(\lambda, \mathcal{F}^{(n)}(\lambda)^{-1}\right) = \mathcal{N}\left(\lambda, \frac{\lambda^2}{n}\right),$$

which leads to the large sample estimate

$$\lambda \in \left[\hat{\lambda}_{\text{ML}} \mp \frac{z_{1-\delta/2} \hat{\lambda}_{\text{ML}}}{\sqrt{n}} \right] \quad \text{with probability } 1 - \delta.$$

The Fisherian picture

The MLE may be biased for finite n , but it is asymptotically unbiased, normal, and efficient.

Akaike information criterion

Recall that, if the model has q free parameters,

$$\text{AIC} = -2 \ell(\mathbf{x}; \hat{\theta}_{\text{ML}}(\mathbf{x})) + 2q.$$

Thus, once the model has been fitted, one may ask:

through which features of the observed data does the AIC depend on the sample?

For the exponential model

Since $q = 1$ and $\hat{\lambda}_{\text{ML}}(\mathbf{x}) = 1/\bar{x}$,

$$\text{AIC}_{\text{Exp}} = 2n(1 + \log \bar{x}) + 2.$$

Hence, for fixed n ,

the data enter the AIC only through $\bar{x}$.

AIC: how the data enter in the normal model

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, with both μ and σ^2 unknown. One has

$$L(\mathbf{x}; (\mu, \sigma^2)) = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2} \implies \hat{\mu}_{\text{ML}}(\mathbf{x}) = \bar{x}, \quad \hat{\sigma}_{\text{ML}}^2(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

At the fitted values,

$$\ell(\mathbf{x}; (\hat{\mu}_{\text{ML}}(\mathbf{x}), \hat{\sigma}_{\text{ML}}^2(\mathbf{x}))) = -\frac{n}{2} [1 + \log(2\pi\hat{\sigma}_{\text{ML}}^2(\mathbf{x}))],$$

and since $q = 2$,

$$\text{AIC}_{\mathcal{N}} = n [1 + \log(2\pi\hat{\sigma}_{\text{ML}}^2(\mathbf{x}))] + 4.$$

How the data enter the AIC

For fixed n ,

the data enter the AIC only through $\hat{\sigma}_{\text{ML}}^2(\mathbf{x})$.

A similar reduction of the data will appear in the Gaussian linear model and will motivate the introduction of the residual vector, whose squared norm measures the part of the data left unexplained after fitting.

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From classical statistics to statistical learning: a probabilistic overview

II — The method of least squares

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The starting point: the data

Consider a collection of observations

$$(\mathbf{x}_i, y_i), \quad i = 1, \dots, n, \quad \text{where} \quad \mathbf{x}_i \in \mathbb{R}^p, \quad y_i \in \mathbb{R}.$$

We organize these data into the *design matrix* and the *response vector*:

$$\mathcal{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} \in \mathbb{R}^{n \times p}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^n.$$

If we wish to include an intercept, we use the augmented design matrix

$$\widetilde{\mathcal{X}} = \begin{pmatrix} 1 & \mathbf{x}_1^\top \\ 1 & \mathbf{x}_2^\top \\ \vdots & \vdots \\ 1 & \mathbf{x}_n^\top \end{pmatrix} \in \mathbb{R}^{n \times (p+1)}.$$

For now

We only have observed data: no probabilistic structure has been introduced.

Linear prediction and residuals

Given a vector of coefficients

$$\beta = (\beta_0, \beta_1, \dots, \beta_p) \in \mathbb{R}^{p+1},$$

we associate to each observation $\mathbf{x}_i$ the linear prediction

$$\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}.$$

In matrix form, the vector of predicted responses is therefore

$$\hat{\mathbf{y}} = \mathcal{X} \beta,$$

where, from now on, we write $\mathcal{X}$ for the augmented design matrix $\widetilde{\mathcal{X}}$.

The corresponding *residual vector* is

$$\mathbf{r}(\beta) = \mathbf{y} - \mathcal{X} \beta,$$

whose i -th component is

$$r_i(\beta) = y_i - (\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}).$$

The basic question

How should we choose β so that the linear predictions $\mathcal{X} \beta$ fit the observed response vector $\mathbf{y}$ as closely as possible?

The residual sum of squares

The observed data define a **cloud of n points** in $\mathbb{R}^{p+1}$, whereas each choice of coefficients $\beta = (\beta_0, \beta_1, \dots, \beta_p) \in \mathbb{R}^{p+1}$ determines the hyperplane

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p.$$

We now have a purely geometric problem: how should we measure the discrepancy between the cloud of points and a given hyperplane?

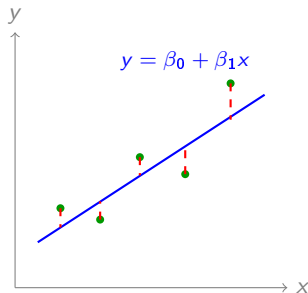
The least squares criterion measures this discrepancy through **the vertical deviations**, by minimizing the corresponding residual sum of squares:

$$\text{RSS}(\beta) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \|\mathbf{y} - \mathcal{X}\beta\|^2,$$

which is a convex quadratic objective.

Thus,

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta \in \mathbb{R}^{p+1}} \text{RSS}(\beta)$$



For $p = 1$, the hyperplane is a line, and the residuals are the vertical deviations from that line.

Solving the least squares problem

We want to minimize

$$\begin{aligned}\text{RSS}(\beta) &= \|\mathbf{y} - \mathcal{X}\beta\|^2 = (\mathbf{y} - \mathcal{X}\beta)^\top (\mathbf{y} - \mathcal{X}\beta) \\ &= \mathbf{y}^\top \mathbf{y} - 2\beta^\top \mathcal{X}^\top \mathbf{y} + \beta^\top \mathcal{X}^\top \mathcal{X} \beta,\end{aligned}$$

so that

$$(\nabla_{\beta} \text{RSS})(\beta) = -2\mathcal{X}^\top \mathbf{y} + 2\mathcal{X}^\top \mathcal{X} \beta \implies (\nabla_{\beta\beta}^2 \text{RSS})(\beta) = 2\mathcal{X}^\top \mathcal{X} \succeq 0.$$

Therefore, RSS is convex. Its stationary points satisfy the *normal equations*

$$\boxed{\mathcal{X}^\top \mathcal{X} \hat{\beta} = \mathcal{X}^\top \mathbf{y},}$$

and every such point is a global minimizer.

If $\mathcal{X}$ has full column rank, $\text{rank}(\mathcal{X}) = p + 1$, then $\mathcal{X}^\top \mathcal{X} \succ 0$ is invertible. In this case, RSS is strictly convex and the minimizer is unique:

$$\boxed{\hat{\beta}_{\text{OLS}} = (\mathcal{X}^\top \mathcal{X})^{-1} \mathcal{X}^\top \mathbf{y}.}$$

A dimensional requirement

Since $\mathcal{X} \in \mathbb{R}^{n \times (p+1)}$, full column rank requires $p + 1 \leq n$.

A second geometric interpretation

The normal equations can be written as

$$\mathcal{X}^\top (\mathbf{y} - \mathcal{X} \hat{\beta}_{\text{OLS}}) = 0.$$

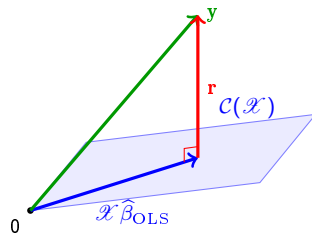
Hence the residual vector $\mathbf{r} = \mathbf{y} - \mathcal{X} \hat{\beta}_{\text{OLS}}$ is orthogonal to the column space $\mathcal{C}(\mathcal{X}) \subset \mathbb{R}^n$.

Thus

$$\mathcal{X} \hat{\beta}_{\text{OLS}} = \Pi_{\mathcal{C}(\mathcal{X})} \mathbf{y}$$

is the orthogonal projection of $\mathbf{y}$ onto $\mathcal{C}(\mathcal{X})$. Equivalently,

$$\mathbf{y} = \mathcal{X} \hat{\beta}_{\text{OLS}} + \mathbf{r}, \quad \mathbf{r} \perp \mathcal{C}(\mathcal{X}).$$



$$\mathcal{X} \hat{\beta}_{\text{OLS}} = \Pi_{\mathcal{C}(\mathcal{X})} \mathbf{y}, \quad \mathbf{r} \perp \mathcal{C}(\mathcal{X}).$$

Geometrically

Least squares chooses, among all vectors in $\mathcal{C}(\mathcal{X})$, the one closest to the observed response vector $\mathbf{y}$.

Injecting probability

So far, $\{\mathbf{x}_i\}$ and $\{y_i\}$ have been treated simply as observed data.

Following the viewpoint adopted in the previous lecture, we now regard them as realizations of random objects:

$$\mathbf{X}_i = (X_{i1}, \dots, X_{ip}) \in \mathbb{R}^p, \quad Y_i \in \mathbb{R}.$$

Thus $\mathbf{X}_i$ is a random vector and Y_i is a random variable.

We now assume that, conditionally on the observed predictors $\mathbf{X}_1 = \mathbf{x}_1, \dots, \mathbf{X}_n = \mathbf{x}_n$,

$$Y_i \mid \mathbf{X}_i = \mathbf{x}_i \sim \mathcal{N}(\beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}, \sigma^2), \quad i = 1, \dots, n,$$

independently.

Equivalently, in matrix form, we have the Gaussian model

$$\mathbf{Y} \mid \mathcal{X} \sim \mathcal{N}_n(\mathcal{X}\beta, \sigma^2 I_n)$$

or

$$\mathbf{Y} = \mathcal{X}\beta + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}_n(0, \sigma^2 I_n).$$

What has changed?

The response vector is now random, while $\beta = (\beta_0, \dots, \beta_p)$ and σ^2 are unknown population parameters. We now have a statistical model a la Fisher with $\Theta = \mathbb{R}^{p+1} \times (0, \infty)$. Hence, $q = p + 2$.

Why least squares?

Under the Gaussian model

$$\mathbf{Y} \mid \mathcal{X} \sim \mathcal{N}_n(\mathcal{X}\beta, \sigma^2 I_n),$$

the likelihood is

$$L(\mathbf{y}; \beta, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \|\mathbf{y} - \mathcal{X}\beta\|^2 \right\},$$

so that the log-likelihood is

$$\ell(\mathbf{y}; \beta, \sigma^2) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \|\mathbf{y} - \mathcal{X}\beta\|^2 = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \text{RSS}(\beta).$$

Therefore, for fixed σ^2 , the log-likelihood is strictly concave in β whenever $\mathcal{X}$ has full column rank. Thus, maximizing the likelihood with respect to β is equivalent to minimizing the residual sum of squares:

$$\boxed{\arg \max_{\beta} L(\mathbf{y}; \beta, \sigma^2) = \arg \min_{\beta} \|\mathbf{y} - \mathcal{X}\beta\|^2} \implies \boxed{\hat{\beta}_{\text{ML}} = \hat{\beta}_{\text{OLS}} = (\mathcal{X}^\top \mathcal{X})^{-1} \mathcal{X}^\top \mathbf{y}.}$$

The statistical meaning of least squares

The same procedure that arose from a purely geometric fitting problem is now the maximum likelihood estimator of the population parameter β under the Gaussian linear model.

Interpretation versus prediction

Under the Gaussian linear model,

$$\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p.$$

The same model can be used with two rather different goals.

Interpretation

We are interested in the population vector β .

Note that

$$\beta_j = \frac{\partial}{\partial x_j} \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}],$$

measures the change in the conditional mean of Y associated with a unit change in x_j , keeping the remaining predictors fixed.

The main object is therefore β .

Prediction

We are interested in the regression function

$$m(\mathbf{x}) = \mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}],$$

especially at a new covariate vector $\mathbf{x}_{\text{new}}$.

Under the fitted linear model,

$$\hat{m}(\mathbf{x}_{\text{new}}) = \hat{\beta}_0 + \hat{\beta}_1 x_{\text{new},1} + \cdots + \hat{\beta}_p x_{\text{new},p}.$$

The main object is therefore $m(\mathbf{x})$.

Inference and prediction

The same fitted model may therefore be viewed either as an inferential object or as a predictive rule.

The geometry of prediction and interpretation

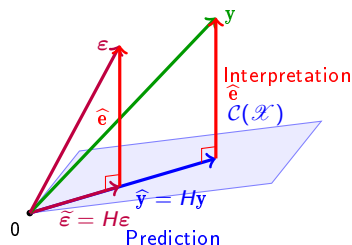
The fitted Gaussian linear model gives statistical meaning to an important orthogonal decomposition in $\mathbb{R}^n$:

$$\mathbf{y} = \hat{\mathbf{y}} + \hat{\mathbf{e}}, \quad \hat{\mathbf{y}} = H\mathbf{y}, \quad \hat{\mathbf{e}} = Q\mathbf{y}, \quad H = \mathcal{X}(\mathcal{X}^\top \mathcal{X})^{-1} \mathcal{X}^\top, \quad Q = I - H.$$

Since $\mathbf{y} - \boldsymbol{\varepsilon} = \mathcal{X}\boldsymbol{\beta} \in \mathcal{C}(\mathcal{X}) \equiv \mathbb{R}^{p+1}$, projection onto the residual space $\mathcal{C}(\mathcal{X})^\perp \equiv \mathbb{R}^{n-p-1}$ gives

$$Q\mathbf{y} = Q\boldsymbol{\varepsilon} = \hat{\mathbf{e}} \implies \hat{\mathbf{e}} \sim \mathcal{N}_{n-p-1}(0, \sigma^2 I_{n-p-1}). \quad (1)$$

Thus $\mathbf{y}$ and $\boldsymbol{\varepsilon}$ have the same projection onto $\mathcal{C}(\mathcal{X})^\perp$, namely, $\hat{\mathbf{e}}$, which is observable from the data.



Predictive component

$\hat{\mathbf{y}} = H\mathbf{y} \in \mathcal{C}(\mathcal{X})$ is the part of $\mathbf{y}$ captured by the fitted linear model and carries its predictive content.

Residual component

Due to (1), the residual $\hat{\mathbf{e}}$ provides the probabilistic starting point for inference and model interpretation.

$$\begin{aligned} \mathbf{y} &= \hat{\mathbf{y}} + \hat{\mathbf{e}}, \\ \boldsymbol{\varepsilon} &= \tilde{\boldsymbol{\varepsilon}} + \hat{\mathbf{e}}. \end{aligned}$$

Since $\text{Cov}(\hat{\mathbf{y}}, \hat{\mathbf{e}}) = \sigma^2 H Q = 0$, and $\hat{\mathbf{y}}$ and $\hat{\mathbf{e}}$ are jointly Gaussian, $\hat{\mathbf{y}} \perp \hat{\mathbf{e}}$, which refines the orthogonal decomposition above.

For the Gaussian linear model,

$$\mathbf{Y} = \mathcal{X}\beta + \epsilon, \quad \epsilon \sim \mathcal{N}_n(0, \sigma^2 I_n),$$

the maximum likelihood estimator of the noise variance σ^2 is

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{n} \|\hat{\mathbf{e}}\|^2.$$

Therefore,

$$\text{AIC} = -2\ell(\hat{\beta}, \hat{\sigma}_{\text{ML}}^2) + 2(p+2) \equiv \underbrace{n \log \left(\frac{\|\hat{\mathbf{e}}\|^2}{n} \right)}_{\text{fit}} + \underbrace{2(p+2)}_{\text{complexity}}$$

Why the residual matters

The previous geometric decomposition showed that $\hat{\mathbf{e}}$ is the observable Gaussian component in $\mathcal{C}(\mathcal{X})^\perp$, and hence the starting point for inference and model interpretation.

AIC adds a complementary fact: likelihood-based goodness of fit depends on the data only through the residual magnitude $\|\hat{\mathbf{e}}\|^2$.

Thus the residual plays a central role both through its *distribution*, for inference, and through its *magnitude*, for model assessment.

From residuals to inference

The residual distribution allows us to estimate the unknown noise level:

$$\hat{\sigma}^2 := \frac{\|\hat{\mathbf{e}}\|^2}{n-p-1}, \quad \Rightarrow \quad \frac{(n-p-1)\hat{\sigma}^2}{\sigma^2} = \frac{\|\hat{\mathbf{e}}\|^2}{\sigma^2} \sim \chi_{n-p-1}^2.$$

Moreover, $\hat{\sigma}^2$ is unbiased and consistent for σ^2 , with $\hat{\beta} \perp\!\!\!\perp \hat{\sigma}^2$.

Now let $\mathbf{S} = (\mathcal{X}^\top \mathcal{X})^{-1}$, $\mathbf{s}_j = \sigma \sqrt{S_{jj}}$, $\hat{\mathbf{s}}_j = \hat{\sigma} \sqrt{S_{jj}}$. Since $(\hat{\beta}_j - \beta_j)/\mathbf{s}_j \sim \mathcal{N}(0, 1)$, we have

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\mathbf{s}}_j} = \frac{\frac{\hat{\beta}_j - \beta_j}{\mathbf{s}_j}}{\sqrt{\frac{\|\hat{\mathbf{e}}\|^2/\sigma^2}{n-p-1}}} \sim t_{n-p-1} \Rightarrow \beta_j \in CI(\beta_j) := \left[\hat{\beta}_j \mp t_{n-p-1, 1-\delta/2} \hat{\mathbf{s}}_j \right] \quad \text{with prob. } 1 - \delta.$$

From inference to interpretation

Confidence intervals quantify which values of each population coefficient β_j are compatible with the data. In particular, $0 \notin CI(\beta_j)$ provides evidence that the predictor x_j contributes to the linear model, conditionally on the other predictors. Hence, the Gaussian linear model is inherently *interpretable*: its coefficients have a direct meaning and their uncertainty can be quantified.

This naturally leads from coefficient interpretation to *variable and model selection*.

Confidence interval for the mean response

For a new (modified) predictor vector $\mathbf{x} = (1, x_1, \dots, x_p) \in \mathbb{R}^{p+1}$, the corresponding population mean response is

$$m(\mathbf{x}) = \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x}) = \mathbf{x}^\top \beta, \quad \mathbf{x} = (x_1, \dots, x_p),$$

and its natural estimator is $\hat{m}(\mathbf{x}) = \mathbf{x}^\top \hat{\beta}$.

Since

$$\hat{\beta} \sim \mathcal{N}_{p+1}(\beta, \sigma^2 S), \quad S = (\mathcal{X}^\top \mathcal{X})^{-1} \implies \mathbf{x}^\top \hat{\beta} \sim \mathcal{N}(\mathbf{x}^\top \beta, \sigma^2 \mathbf{x}^\top S \mathbf{x}),$$

we obtain, with confidence level $1 - \delta$,

$$\frac{\mathbf{x}^\top \hat{\beta} - \mathbf{x}^\top \beta}{\hat{\sigma} \sqrt{\mathbf{x}^\top S \mathbf{x}}} \sim t_{n-p-1} \implies m(\mathbf{x}) = \mathbf{x}^\top \beta \in \left[\mathbf{x}^\top \hat{\beta} \mp t_{n-p-1, 1-\delta/2} \hat{\sigma} \sqrt{\mathbf{x}^\top S \mathbf{x}} \right].$$

What is being estimated?

The target is the *mean response* $m(\mathbf{x}) = \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x})$, not a future individual response: its uncertainty comes only from estimating the unknown regression coefficients.

Prediction interval for a new response

For the same new predictor vector $\mathbf{x} = (1, x_1, \dots, x_p)$, consider now a new individual response

$$Y_{\mathbf{x}} = \mathbf{x}^{\top} \beta + \varepsilon_{\mathbf{x}}, \quad \varepsilon_{\mathbf{x}} \sim \mathcal{N}(0, \sigma^2), \quad \varepsilon_{\mathbf{x}} \perp\!\!\!\perp \{\text{training data}\}.$$

The natural predictor remains $\hat{Y}_{\mathbf{x}} = \mathbf{x}^{\top} \hat{\beta} = \hat{m}(\mathbf{x})$, but

$$Y_{\mathbf{x}} - \mathbf{x}^{\top} \hat{\beta} = \varepsilon_{\mathbf{x}} + \mathbf{x}^{\top} (\beta - \hat{\beta}) \implies Y_{\mathbf{x}} - \mathbf{x}^{\top} \hat{\beta} \sim \mathcal{N} \left(0, \sigma^2 \left(1 + \mathbf{x}^{\top} S_{\mathbf{X}} \mathbf{x} \right) \right),$$

and therefore, again with probability $1 - \delta$,

$$\frac{Y_{\mathbf{x}} - \mathbf{x}^{\top} \hat{\beta}}{\hat{\sigma} \sqrt{1 + \mathbf{x}^{\top} S_{\mathbf{X}} \mathbf{x}}} \sim t_{n-p-1} \implies Y_{\mathbf{x}} \in \left[\mathbf{x}^{\top} \hat{\beta} \mp t_{n-p-1, 1-\delta/2} \hat{\sigma} \sqrt{1 + \mathbf{x}^{\top} S_{\mathbf{X}} \mathbf{x}} \right]$$

Mean response versus individual prediction

Both problems use the same point predictor $\mathbf{x}^{\top} \hat{\beta}$, but their uncertainty differs:

$$\underbrace{\hat{\sigma} \sqrt{\mathbf{x}^{\top} S_{\mathbf{X}} \mathbf{x}}}_{\text{uncertainty about the mean}} \longrightarrow \underbrace{\hat{\sigma} \sqrt{1 + \mathbf{x}^{\top} S_{\mathbf{X}} \mathbf{x}}}_{\text{prediction uncertainty}}.$$

The extra 1 reflects the variance of the new noise term $\varepsilon_{\mathbf{x}}$, which remains even under perfect knowledge of $m(\mathbf{x})$.

From explanation to prediction: two cultures

For much of classical statistics, the main goal was to specify a probabilistic model for the data-generating mechanism and then infer its unknown parameters.

Data modeling culture

data $\longrightarrow$ probabilistic model $\longrightarrow$ parameters

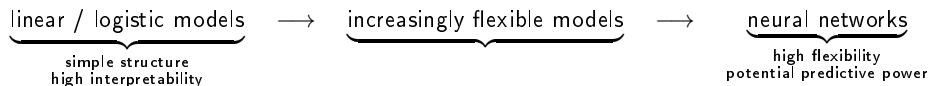
inference, uncertainty, interpretation

Algorithmic modeling culture

data $\longrightarrow$ learned predictor $\longrightarrow$ prediction

predictive accuracy, flexibility

Breiman's influential 2001 essay *Statistical Modeling: The Two Cultures* argued that statistics should move beyond an almost exclusive reliance on stochastic data models and give a central role to predictive performance.



A change of emphasis

Modern statistical learning increasingly asks not only *what does the model tell us about the data-generating mechanism?*, but also *how well can we predict new observations?*

Logistic regression: a probabilistic classifier

Suppose now that the response is binary, $Y \in \{0, 1\}$, so that for a predictor vector $\mathbf{x} = (x_1, \dots, x_p)$, the quantity of interest is the conditional probability

$$p(\mathbf{x}) = \mathbb{P}(Y = 1 \mid \mathbf{X} = \mathbf{x}), \quad \mathbf{x} = (x_1, \dots, x_p).$$

Since a linear model $p(\mathbf{x}) = \mathbf{x}^\top \beta$, $\mathbf{x} = (1, x_1, \dots, x_p)$, is generally inappropriate, because its values need not remain in $[0, 1]$, logistic regression instead assumes that the *log-odds* are linear:

$$\text{logit } p(\mathbf{x}) := \log \frac{p(\mathbf{x})}{1 - p(\mathbf{x})} = \mathbf{x}^\top \beta \iff p(\mathbf{x}) = \sigma(\mathbf{x}^\top \beta), \quad \sigma(t) = \frac{1}{1 + e^{-t}},$$

where $\sigma = \text{logit}^{-1}$ is the output sigmoid.

The key structural feature

The probability $p(\mathbf{x})$ is a nonlinear function of $\mathbf{x}$, but its *linear predictor*

$$\eta(\mathbf{x}) = \mathbf{x}^\top \beta$$

is still linear in the covariates.

Logistic regression is still interpretable

The logistic model

$$\text{logit } p(\mathbf{x}) = \mathbf{x}^\top \beta,$$

retains a direct interpretation of its coefficients: the shift $x_j \rightarrow x_j + 1$, keeping the other predictors fixed, implies

$$\text{logit } p(\mathbf{x}) \rightarrow \text{logit } p(\mathbf{x}) + \beta_j \iff \text{the odds are multiplied by } e^{\beta_j}.$$

Logistic regression as a model: the conditional binary response follows a Bernoulli distribution

Writing

$$p_i = \sigma(\mathbf{x}_i^\top \beta) = \frac{1}{1 + e^{-\mathbf{x}_i^\top \beta}}, \quad Y_i \mid \mathbf{X}_i = \mathbf{x}_i \sim \text{Bernoulli}(p_i), \quad (2)$$

the likelihood is

$$L(\mathbf{y}; \beta) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i} \implies \ell(\mathbf{y}; \beta) = \sum_{i=1}^n \left[y_i \mathbf{x}_i^\top \beta - \log(1 + e^{\mathbf{x}_i^\top \beta}) \right],$$

so that

$$\nabla_{\beta} \ell = \mathcal{X}^\top (\mathbf{y} - \mathbf{p}), \quad \nabla_{\beta\beta}^2 \ell = -\mathcal{X}^\top W(\beta) \mathcal{X}, \quad W(\beta) = \text{diag}(p_i(1 - p_i)).$$

From the computations above, for full column rank $\mathcal{X}$ we see that:

- the log-likelihood is strictly concave;
- the MLE satisfies $\mathcal{X}^\top (\mathbf{y} - \boldsymbol{\mu}) = 0$, which is nonlinear in β due to (2);
- the Fisher information matrix is $\mathcal{F}(\beta) = \mathcal{X}^\top W(\beta) \mathcal{X}$.

Maximum likelihood estimation gives, asymptotically,

$$\hat{\beta} \approx \mathcal{N}_{p+1}(\beta, \mathcal{F}(\beta)^{-1}) = \mathcal{N}_{p+1}\left(\beta, (\mathcal{X}^\top W(\beta) \mathcal{X})^{-1}\right),$$

so that confidence intervals for the coefficients remain available.

Recall that, for the Gaussian linear model,

$$\hat{\beta} \sim \mathcal{N}_{p+1}\left(\beta, \sigma^2 (\mathcal{X}^\top \mathcal{X})^{-1}\right),$$

exactly, for finite n .

Still within the classical paradigm

Logistic regression preserves

meaningful parameters + inference + interpretability.

Unlike the Gaussian linear model, inference is now asymptotic: exact orthogonal projection is replaced by a local, likelihood-based geometry encoded by the β -dependent Fisher information $\mathcal{X}^\top W(\beta) \mathcal{X}$.

From logistic regression to a neural network

Recall that logistic regression models the conditional probability as

$$p(\mathbf{x}) = \mathbb{P}(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \sigma(\beta_0 + \beta_{\bullet}^{\top} \mathbf{x}), \quad \beta_{\bullet} = (\beta_1, \dots, \beta_p),$$

A simple neural network replaces the linear predictor by a learned nonlinear representation:

$$h_k(\mathbf{x}) = \phi(\mathbf{w}_{k0} + \mathbf{w}_k^{\top} \mathbf{x}), \quad k = 1, \dots, m, \quad f_{\theta}(\mathbf{x}) = \alpha_0 + \sum_{k=1}^m \alpha_k h_k(\mathbf{x}),$$

where $\mathbf{w}_k \in \mathbb{R}^p$, $\mathbf{w}_{k0}, \alpha_k \in \mathbb{R}$, $\theta = (\alpha_0, (\alpha_k, \mathbf{w}_{k0}, \mathbf{w}_k)_{k=1}^m) \in \mathbb{R}^{1+m(p+2)}$ collects all network parameters, and ϕ is a nonlinear activation function. The output probability is then

$$p_{\theta}(\mathbf{x}) = \sigma(f_{\theta}(\mathbf{x})) = \sigma\left(\alpha_0 + \sum_{k=1}^m \alpha_k \phi(\mathbf{w}_{k0} + \mathbf{w}_k^{\top} \mathbf{x})\right).$$

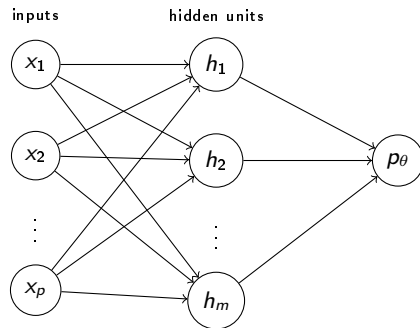
The essential transition

linear predictor $\longrightarrow$ learned nonlinear representation.

Logistic regression is recovered when this nonlinear hidden representation is removed.

The transition from logistic regression to a neural network changes the class of admissible predictors:

$$\mathcal{F}_{\text{logit}} = \left\{ \mathbf{x} \mapsto \sigma\left(\beta_0 + \beta_{\bullet}^{\top} \mathbf{x}\right) \right\} \longrightarrow \mathcal{F}_{\text{NN}}^{(m, \phi)} = \left\{ \mathbf{x} \mapsto \sigma\left(\alpha_0 + \sum_{k=1}^m \alpha_k \phi(w_{k0} + w_k^{\top} \mathbf{x})\right) \right\}.$$



A change of hypothesis class

The hidden-layer width m and the activation function ϕ determine the family of nonlinear functions available to the model.

The number of parameters also grows from

$$p + 1 \quad \text{to} \quad m(p + 2) + 1.$$

Increasing flexibility therefore means changing the hypothesis class itself.

The probabilistic model behind the network

For binary responses, the neural network may still be viewed as a statistical model:

$$Y_i \mid \mathbf{X}_i = \mathbf{x}_i \sim \text{Bernoulli}(p_\theta(\mathbf{x}_i)),$$

where

$$p_\theta(\mathbf{x}) = \sigma(f_\theta(\mathbf{x})), \quad f_\theta(\mathbf{x}) = \alpha_0 + \sum_{k=1}^m \alpha_k \phi(\mathbf{w}_{k0} + \mathbf{w}_k^\top \mathbf{x}), \quad \theta = (\alpha_0, (\alpha_k, \mathbf{w}_{k0}, \mathbf{w}_k)_{k=1}^m) \in \mathbb{R}^{1+m(p+2)}.$$

Thus, for independent observations,

$$L(\theta) = \prod_{i=1}^n p_\theta(\mathbf{x}_i)^{y_i} (1 - p_\theta(\mathbf{x}_i))^{1-y_i} \implies -\ell(\theta) = \sum_{i=1}^n \underbrace{[-y_i \log p_\theta(\mathbf{x}_i) - (1 - y_i) \log(1 - p_\theta(\mathbf{x}_i))]}_{\text{binary cross-entropy loss}}.$$

What has changed?

The conditional distribution is still Bernoulli, and the fitting criterion is still likelihood-based. What has changed substantially is the model for the conditional probability:

$$\sigma(\text{linear predictor}) \longrightarrow \sigma(\text{learned nonlinear predictor}).$$

The binary cross-entropy loss is therefore simply the negative Bernoulli log-likelihood.

The estimator is still defined by maximum likelihood:

$$\hat{\theta} = \arg \max_{\theta} L(\theta) = \arg \min_{\theta} [-\ell(\theta)].$$

For logistic regression, the negative log-likelihood is **convex** in β , so the optimization problem has a particularly favorable structure. For a neural network, however,

$$p_{\theta}(\mathbf{x}) = \sigma \left(\alpha_0 + \sum_{k=1}^m \alpha_k \phi(w_{k0} + w_k^{\top} \mathbf{x}) \right)$$

has a layered nonlinear dependence on the parameters θ . The resulting objective is generally **nonconvex**. A basic optimization scheme is gradient descent:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_{\theta} [-\ell(\theta^{(t)})],$$

where $\eta > 0$ is the learning rate. **Backpropagation** provides an efficient way of computing $\nabla_{\theta} [-\ell(\theta)]$ through repeated applications of the chain rule.

Same statistical principle, harder optimization

The probabilistic principle has not changed: fit the model by maximum likelihood. What has changed is the geometry of the optimization problem: **convex** likelihood optimization is replaced by a generally **nonconvex** search over a much richer parameter space.

What have we gained — and what have we lost?

Consider the two models for the conditional probability:

Logistic regression

- $p_{\beta}(\mathbf{x}) = \sigma(\beta_0 + \beta_{\bullet}^T \mathbf{x})$;
- simple, rigid model class;
- few parameters;
- coefficients have direct meaning;
- limited ability to represent nonlinear structure;
- classical inference remains available.

Neural network

- $p_{\theta}(\mathbf{x}) = \sigma(\alpha_0 + \sum_{k=1}^m \alpha_k \phi(w_{k0} + w_k^T \mathbf{x}))$;
- much richer nonlinear model class;
- many more parameters;
- individual parameters lose direct interpretation;
- greater representational flexibility;
- classical parameter-by-parameter inference becomes much less natural.

The interpretation–prediction trade-off

Greater flexibility may improve predictive performance when the underlying structure is sufficiently complex, but this comes at the cost of a less transparent model:

interpretability $\longleftrightarrow$ flexibility / potential predictive power.

More flexibility, however, does not by itself guarantee better prediction.

From parametric models to function classes

So far, each model has been described through a finite-dimensional parameter vector, with each parameter value determining a function of the predictors.

Thus, we may equivalently describe each model by the corresponding class of predictor functions:

Linear regression

$$\mathcal{F}_{\text{lin}} = \left\{ \beta_0 + \beta_{\bullet}^{\top} \mathbf{x} \right\}$$

Logistic regression

$$\mathcal{F}_{\text{logit}} = \left\{ \sigma \left(\beta_0 + \beta_{\bullet}^{\top} \mathbf{x} \right) \right\}$$

Neural network

$$\mathcal{F}_{\text{NN}}^{(m, \phi)} = \left\{ \sigma \left(\alpha_0 + \sum_{k=1}^m \alpha_k \phi(w_{k0} + w_k^{\top} \mathbf{x}) \right) \right\}.$$

A change of viewpoint

The emphasis now shifts from estimating a particular parameter to selecting a function

$$f \in \mathcal{F}.$$

This function-class viewpoint includes fixed-dimensional parametric models, but also opens the way to genuinely nonparametric learning.

The (supervised) statistical learning problem

Given an iid sample $(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n) \sim P_{(\mathbf{X}, Y)}$, with $P_{(\mathbf{X}, Y)}$ unknown, a class of functions $\mathcal{F} = \{f : \mathcal{X} \rightarrow \mathcal{A}\}$ and a loss function $\ell : \mathcal{A} \times \mathcal{Y} \rightarrow [0, \infty)$, we may consider the *empirical risk*

$$\hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{X}_i), Y_i),$$

so that statistical learning selects

$$\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{R}_n(f).$$

Clearly, the models considered so far arise from particular choices of $\mathcal{F}$ and ℓ . The quantity we would really like to minimize, however, is the *population risk*

$$R(f) = \mathbb{E}_{P_{(\mathbf{X}, Y)}}(\ell(f(\mathbf{X}), Y)),$$

which governs predictive performance. This tension gives rise to the central problem in statistical learning.

The central question left open: when does empirical success generalize?

In other words: *is the generalization gap $|R(\hat{f}) - \hat{R}_n(\hat{f})|$ small?* A standard way to guarantee this is to control $|R(f) - \hat{R}_n(f)|$ uniformly over $f \in \mathcal{F}$, a task whose difficulty depends on the “complexity” of the hypothesis class $\mathcal{F}$. Notice that interpretability does not appear in this formulation: the focus is on prediction and generalization. In this sense, Breiman’s predictive culture became the dominant language of modern statistical learning.

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From classical statistics to statistical learning: a probabilistic overview

III — Generalization and model complexity

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ConfIA — FAPEPI

The central problem of (supervised) statistical learning revisited

Let

$$(\mathbf{X}_1, Y_1), \dots, (\mathbf{X}_n, Y_n) \stackrel{\text{iid}}{\sim} P_{(\mathbf{X}, Y)}, \quad (\mathbf{X}, Y) \in \mathcal{X} \times \mathcal{Y},$$

where the distribution $P_{(\mathbf{X}, Y)}$ is unknown.

Given a hypothesis class $\mathcal{F} \subset \{f : \mathcal{X} \rightarrow \mathcal{Y}\}$ and a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow [0, \infty)$, define

$$\text{population risk : } R(f) = \mathbb{E}[\ell(f(\mathbf{X}), Y)] \quad \text{and} \quad \text{empirical risk : } \hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{X}_i), Y_i).$$

Assuming the minimizers are unique, there are now two distinguished elements of $\mathcal{F}$:

$$\boxed{f_{\mathcal{F}}^* = \arg \min_{f \in \mathcal{F}} R(f)} \quad \text{and} \quad \boxed{\hat{f}_n = \arg \min_{f \in \mathcal{F}} \hat{R}_n(f)}.$$

- $f_{\mathcal{F}}^*$ is the best predictor available within the class $\mathcal{F}$, but depends on the unknown population law;
- $\hat{f}_n$ is computable from the observed sample, but is itself random because it depends on the data.

The central question

Under what conditions does empirical risk minimization generalize? That is, when can we guarantee that

$$\boxed{R(\hat{f}_n) - R(f_{\mathcal{F}}^*) \longrightarrow 0} \quad \text{as } n \rightarrow \infty?$$

Why the ordinary law of large numbers is not enough

We have

$$0 \leq R(\hat{f}_n) - R(f_{\mathcal{F}}^*) = \underbrace{R(\hat{f}_n) - \hat{R}_n(\hat{f}_n)}_{\text{generalization gap}} + \underbrace{\hat{R}_n(\hat{f}_n) - \hat{R}_n(f_{\mathcal{F}}^*)}_{\leq 0} + \underbrace{\hat{R}_n(f_{\mathcal{F}}^*) - R(f_{\mathcal{F}}^*)}_{\xrightarrow{P} 0 \text{ by LLN}},$$

where we used the ordinary LLN in the last term: for each **fixed** $f \in \mathcal{F}$,

$$\hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(\mathbf{X}_i), Y_i) \xrightarrow{P} \mathbb{E}[\ell(f(\mathbf{X}), Y)] = R(f).$$

However, we are **not** allowed to use this with $f = \hat{f}_n$ because the empirical minimizer $\hat{f}_n = \arg \min_{f \in \mathcal{F}} \hat{R}_n(f)$ is **not fixed**: it is selected after observing the data.

Consequently, we are left with the rather crude estimate

$$R(\hat{f}_n) - R(f_{\mathcal{F}}^*) \leq R(\hat{f}_n) - \hat{R}_n(\hat{f}_n) + o_P(1) \leq \sup_{f \in \mathcal{F}} |\hat{R}_n(f) - R(f)| + o_P(1).$$

The new probabilistic problem

We need a **uniform law of large numbers** over the entire hypothesis class:

$$\sup_{f \in \mathcal{F}} |\hat{R}_n(f) - R(f)| \xrightarrow{P} 0.$$

From a uniform LLN to an empirical process problem

Let $\mathbf{Z} = (\mathbf{X}, Y) \in \mathcal{Z} := \mathcal{X} \times \mathcal{Y}$, and define $f \in \mathcal{F} \mapsto g_f : \mathcal{Z} \rightarrow [0, \infty)$ by $g_f(\beta_{\bullet}^{\top} \mathbf{x}, y) = \ell(f(\mathbf{x}), y)$. Thus, $\mathcal{F}$ induces the loss class $\mathcal{G}_{\ell} := \{g_f : f \in \mathcal{F}\}$. Now, for any measurable $g : \mathcal{Z} \rightarrow \mathbb{R}$, write

$$Pg := \mathbb{E}[g(\mathbf{Z})], \quad P_n g := \frac{1}{n} \sum_{i=1}^n g(\mathbf{Z}_i) \quad \implies \quad R(f) = Pg_f, \quad \hat{R}_n(f) = P_n g_f,$$

so the quantity controlling generalization becomes

$$\sup_{g \in \mathcal{G}_{\ell}} |(P_n - P)g|.$$

Pointwise versus uniform convergence

For every fixed $g \in \mathcal{G}_{\ell}$, the ordinary LLN gives $P_n g - Pg \xrightarrow{P} 0$, but statistical learning requires much more:

$$\sup_{g \in \mathcal{G}_{\ell}} |P_n g - Pg| \xrightarrow{P} 0.$$

The question has changed

Which properties of the class $\mathcal{G}_{\ell}$, and hence of $\mathcal{F}$, make this uniform convergence possible?

Symmetrization I: introducing a ghost sample

Let $\mathbf{Z}_1, \dots, \mathbf{Z}_n \stackrel{\text{iid}}{\sim} P$ as above and let $\mathbf{Z}'_1, \dots, \mathbf{Z}'_n$ be an independent copy of the sample. Define

$$P'_n g = \frac{1}{n} \sum_{i=1}^n g(\mathbf{Z}'_i).$$

If $\mathcal{G} = \sigma(\mathbf{Z}_1, \dots, \mathbf{Z}_n)$, then, for every $g \in \mathcal{G}_\ell$,

$$\mathbb{E}[P'_n g \mid \mathcal{G}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[g(\mathbf{Z}'_i) \mid \mathcal{G}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[g(\mathbf{Z}'_i)] = P g, \quad \mathbb{E}[P_n g \mid \mathcal{G}] = P_n g.$$

Hence

$$\sup_{g \in \mathcal{G}_\ell} |P_n g - P g| = \sup_{g \in \mathcal{G}_\ell} |\mathbb{E}[P_n g - P'_n g \mid \mathcal{G}]| \leq \mathbb{E} \left[\sup_{g \in \mathcal{G}_\ell} |P_n g - P'_n g| \mid \mathcal{G} \right],$$

by conditional Jensen. Therefore,

$$\mathbb{E} \sup_{g \in \mathcal{G}_\ell} |P_n g - P g| \leq \mathbb{E} \sup_{g \in \mathcal{G}_\ell} |P_n g - P'_n g|.$$

The key idea

The unknown population mean $P g$ has been replaced by the empirical mean $P'_n g$ of the ghost sample.

Symmetrization II: Rademacher signs

If $\varepsilon_1, \dots, \varepsilon_n$ are independent Rademacher random variables, independent of both samples $\mathbf{Z}$ and $\mathbf{Z}'$,

$$g(\mathbf{Z}_i) - g(\mathbf{Z}'_i) \text{ is symmetric} \implies \mathbb{E} \sup_{g \in \mathcal{G}_\ell} \left| \frac{1}{n} \sum_{i=1}^n (g(\mathbf{Z}_i) - g(\mathbf{Z}'_i)) \right| = \mathbb{E} \sup_{g \in \mathcal{G}_\ell} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (g(\mathbf{Z}_i) - g(\mathbf{Z}'_i)) \right|.$$

By the triangle inequality,

$$\mathbb{E} \sup_{g \in \mathcal{G}_\ell} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i (g(\mathbf{Z}_i) - g(\mathbf{Z}'_i)) \right| \leq \mathbb{E} \sup_{g \in \mathcal{G}_\ell} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(\mathbf{Z}_i) \right| + \mathbb{E} \sup_{g \in \mathcal{G}_\ell} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(\mathbf{Z}'_i) \right|.$$

Since the two samples have the same distribution,

$$\boxed{\mathbb{E} \sup_{g \in \mathcal{G}_\ell} |P_n g - P g| \leq 2 \mathfrak{R}_n(\mathcal{G}_\ell)} \quad \text{[Symmetrization bound]}$$

where

$$\boxed{\mathfrak{R}_n(\mathcal{G}_\ell) := \mathbb{E} \sup_{g \in \mathcal{G}_\ell} \left| \frac{1}{n} \sum_{i=1}^n \varepsilon_i g(\mathbf{Z}_i) \right|} \quad \text{[Rademacher complexity of } \mathcal{G}_\ell \text{].}$$

Interpretation

$\mathfrak{R}_n(\mathcal{G}_\ell)$ measures how well the class can adapt to purely random sign fluctuations in the sample.

From expectation to high probability

Assume that the induced losses are uniformly bounded, $0 \leq g(\mathbf{z}) \leq M, g \in \mathcal{G}_\ell, \mathbf{z} \in \mathcal{Z}$, and consider the random variable

$$\Phi(\mathbf{Z}_1, \dots, \mathbf{Z}_n) := \sup_{g \in \mathcal{G}_\ell} |P_n g - P g|.$$

If one observation $\mathbf{Z}_i$ is replaced by $\mathbf{Z}'_i$, then

$$|\Phi(\mathbf{Z}_1, \dots, \mathbf{Z}_i, \dots, \mathbf{Z}_n) - \Phi(\mathbf{Z}_1, \dots, \mathbf{Z}'_i, \dots, \mathbf{Z}_n)| \leq \frac{M}{n}. \quad [\text{Bounded differences}]$$

Hence, by McDiarmid's concentration inequality, with probability at least $1 - \delta$,

$$\Phi \leq \mathbb{E}\Phi + M \sqrt{\frac{\log(1/\delta)}{2n}}.$$

From the symmetrization bound, $\mathbb{E}\Phi \leq 2\mathfrak{R}_n(\mathcal{G}_\ell)$, and therefore, again with probability at least $1 - \delta$,

$$\sup_{g \in \mathcal{G}_\ell} |P_n g - P g| \leq 2\mathfrak{R}_n(\mathcal{G}_\ell) + M \sqrt{\frac{\log(1/\delta)}{2n}}.$$

Uniform law of large numbers

If $\mathfrak{R}_n(\mathcal{G}_\ell) \xrightarrow{P} 0$, then $\sup_{g \in \mathcal{G}_\ell} |P_n g - P g| \xrightarrow{P} 0$.

From generalization to model selection

So far, the hypothesis class $\mathcal{F}$ has been regarded as fixed. Now let $\mathcal{F} \subset \mathcal{F}_{\text{all}}$, where $\mathcal{F}_{\text{all}}$ is a sufficiently rich ambient class of admissible predictors, and define

$$f^* = \arg \min_{f \in \mathcal{F}_{\text{all}}} R(f).$$

The total excess risk admits the decomposition

$$R(\hat{f}_n) - R(f^*) = \underbrace{R(f_{\mathcal{F}}^*) - R(f^*)}_{\text{approximation error}} + \underbrace{R(\hat{f}_n) - R(f_{\mathcal{F}}^*)}_{\text{estimation error}},$$

where, as before, $f_{\mathcal{F}}^*$ and $\hat{f}_n$ are the population and empirical minimizers within $\mathcal{F}$, respectively.

- The **approximation error** measures the price paid for restricting the predictor to the class $\mathcal{F}$.
- The **estimation error** measures the price paid for learning, within $\mathcal{F}$, from a finite random sample: it has been controlled above through uniform convergence and Rademacher complexity.

The new question

How rich should the hypothesis class $\mathcal{F}$ be?

Squared loss: approximation as bias

In the regression setting with squared loss, $\ell(f(\mathbf{x}), y) = (y - f(\mathbf{x}))^2$, the optimal predictor in the ambient class is

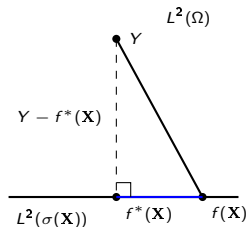
$$f^*(\mathbf{x}) = \mathbb{E}(Y \mid \mathbf{X} = \mathbf{x}).$$

Since $f^*(\mathbf{X})$ is the orthogonal projection of Y onto $L^2(\sigma(\mathbf{X}))$, the space of predictors depending on $\mathbf{X}$, naturally identified with $L^2(P_{\mathbf{X}})$, for any $f \in \mathcal{F}_{\text{all}}$ we have

$$R(f) - R(f^*) = \mathbb{E}_{P_{\mathbf{X}}} \left[(f(\mathbf{X}) - f^*(\mathbf{X}))^2 \right].$$

Therefore,

$$R(f_{\mathcal{F}}^*) - R(f^*) = \|f_{\mathcal{F}}^* - f^*\|_{L^2(P_{\mathbf{X}})}^2.$$



Interpretation

The approximation error measures the squared $L^2(P_{\mathbf{X}})$ -distance between the optimal predictor f^* and its best approximation $f_{\mathcal{F}}^*$ within $\mathcal{F}$. Thus, restricting the predictor to $\mathcal{F}$ produces an **integrated squared bias**.

Squared loss: estimation as variance

Still in the regression setting with squared loss, the same orthogonality argument within $\mathcal{F}$ gives

$$R(\hat{f}_n) - R(f_{\mathcal{F}}^*) = \mathbb{E}_{P_{\mathbf{X}}} \left[(\hat{f}_n(\mathbf{X}) - f_{\mathcal{F}}^*(\mathbf{X}))^2 \right].$$

Unlike the approximation error, this quantity is **data-dependent**, since $\hat{f}_n$ depends on the training sample $\mathbf{Z}$.

Taking a further expectation with respect to the training sample,

$$\mathbb{E}_{\mathbf{Z}} \left[R(\hat{f}_n) - R(f_{\mathcal{F}}^*) \right] = \mathbb{E}_{P_{\mathbf{X}}} \left[\mathbb{V}_{\mathbf{Z}}(\hat{f}_n(\mathbf{X})) \right].$$

Interpretation

The estimation error reflects the variability of the learned predictor across different random training samples. It is therefore the integrated **variance contribution**.

Bias–variance in statistical learning

Under squared loss, we thus recover the classical bias–variance trade-off in a new form:

approximation error $\longleftrightarrow$ bias, estimation error $\longleftrightarrow$ variance.

The complexity trade-off

As the hypothesis class becomes richer, $\mathcal{F} \nearrow \mathcal{F}_{\text{all}}$, two opposite effects typically occur:

Approximation

A richer class can approximate the optimal predictor f^* more closely:

$R(f_{\mathcal{F}}^*) - R(f^*)$ tends to decrease.

$\mathcal{F} \nearrow \mathcal{F}_{\text{all}} \implies$ approximation error $\downarrow$

Estimation

A richer class is more flexible and therefore harder to control uniformly from a finite sample:

$R(\hat{f}_n) - R(f_{\mathcal{F}}^*)$ becomes harder to control.

$\mathcal{F} \nearrow \mathcal{F}_{\text{all}} \implies$ complexity $\uparrow$

restricted $\mathcal{F}$
larger approximation error
easier estimation
underfitting

rich $\mathcal{F}$
smaller approximation error
harder estimation
overfitting

Model selection

The choice of $\mathcal{F}$ amounts to balancing **approximation** against **estimation**: complexity is simultaneously a source of expressive power and a potential obstruction to generalization.

From loss complexity to model complexity

The generalization bound obtained above involves the induced loss class

$$\mathcal{G}_\ell = \{(\mathbf{x}, y) \mapsto \ell(f(\mathbf{x}), y) : f \in \mathcal{F}\}.$$

For a given sample $\mathbf{X}_1, \dots, \mathbf{X}_n$, define the **empirical Rademacher complexity**

$$\hat{\mathfrak{R}}_n(\mathcal{F}) = \mathbb{E}_\epsilon \left[\sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \epsilon_i f(\mathbf{X}_i) \right| \middle| \mathbf{X}_1, \dots, \mathbf{X}_n \right].$$

Its expectation over the sample gives

$$\mathfrak{R}_n(\mathcal{F}) = \mathbb{E}_{\mathbf{X}_1, \dots, \mathbf{X}_n} [\hat{\mathfrak{R}}_n(\mathcal{F})].$$

Contraction principle

If $t \mapsto \ell(t, y)$ is L_ℓ -Lipschitz, uniformly in y , then, up to universal constants,

$$\hat{\mathfrak{R}}_n(\mathcal{G}_\ell) \lesssim L_\ell \hat{\mathfrak{R}}_n(\mathcal{F}), \quad \mathfrak{R}_n(\mathcal{G}_\ell) \lesssim L_\ell \mathfrak{R}_n(\mathcal{F}).$$

The practical problem

Generalization can therefore be studied by estimating the complexity of the **predictor class** $\mathcal{F}$. How does this complexity behave for concrete statistical models?

Binary classification: from patterns to Rademacher complexity

Let $\mathcal{F} \subset \{0, 1\}^{\mathcal{X}}$ be a class of binary classifiers, and fix a sample $\mathbf{X}_1, \dots, \mathbf{X}_n$. On this sample, each $f \in \mathcal{F}$ produces a binary vector $(f(\mathbf{X}_1), \dots, f(\mathbf{X}_n)) \in \{0, 1\}^n$. Thus $\mathcal{F}$ induces the finite set of labeling patterns

$$\mathcal{V}_{\mathcal{F}}(\mathbf{X}_1, \dots, \mathbf{X}_n) = \{(f(\mathbf{X}_1), \dots, f(\mathbf{X}_n)) : f \in \mathcal{F}\}.$$

Massart's finite-class lemma

Since the empirical Rademacher complexity only depends on these vectors,

$$\hat{\mathfrak{R}}_n(\mathcal{F}) \lesssim \sqrt{\frac{\log |\mathcal{V}_{\mathcal{F}}(\mathbf{X}_1, \dots, \mathbf{X}_n)|}{n}} \implies \boxed{\hat{\mathfrak{R}}_n(\mathcal{F}) \lesssim \sqrt{\frac{\log \Pi_{\mathcal{F}}(n)}{n}}},$$

where

$$\Pi_{\mathcal{F}}(n) = \sup_{\mathbf{x}_1, \dots, \mathbf{x}_n} |\mathcal{V}_{\mathcal{F}}(\mathbf{x}_1, \dots, \mathbf{x}_n)|$$

is the **growth function** of $\mathcal{F}$.

Interpretation

For binary classification, Rademacher complexity is controlled by the number of distinct labeling patterns that the class can realize on a finite sample.

VC dimension: worst-case labeling capacity

Recall that, for points $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathcal{X}$,

$$\mathcal{V}_{\mathcal{F}}(\mathbf{x}_1, \dots, \mathbf{x}_m) = \{ (f(\mathbf{x}_1), \dots, f(\mathbf{x}_m)) : f \in \mathcal{F} \}.$$

Shattering

The set

$$\{\mathbf{x}_1, \dots, \mathbf{x}_m\}$$

is said to be **shattered** by $\mathcal{F}$ if every possible binary labeling can be realized, that is,

$$|\mathcal{V}_{\mathcal{F}}(\mathbf{x}_1, \dots, \mathbf{x}_m)| = 2^m.$$

The **VC dimension** of $\mathcal{F}$ is

$$h_{\text{VC}} = \text{VCdim}(\mathcal{F}) = \sup \{m : \text{some set of } m \text{ points is shattered by } \mathcal{F}\}.$$

Interpretation

VC dimension measures the size of the largest sample on which $\mathcal{F}$ can realize *every* possible binary labeling. It is therefore a worst-case combinatorial measure of capacity.

VC dimension controls the growth function

The VC dimension identifies the largest sample that can be completely shattered by $\mathcal{F}$. What can be said about the number of labeling patterns beyond this threshold?

Sauer–Shelah lemma

If $n \geq h_{\text{VC}} = \text{VCdim}(\mathcal{F})$, then

$$\Pi_{\mathcal{F}}(n) \leq \sum_{j=0}^{h_{\text{VC}}} \binom{n}{j} \leq \left(\frac{en}{h_{\text{VC}}} \right)^{h_{\text{VC}}}.$$

Combining this with Massart's bound gives

$$\hat{\mathfrak{R}}_n(\mathcal{F}) \lesssim \sqrt{\frac{h_{\text{VC}} \log(en/h_{\text{VC}})}{n}}.$$

Therefore,

$$h_{\text{VC}} \implies \Pi_{\mathcal{F}}(n) \implies \hat{\mathfrak{R}}_n(\mathcal{F}) \implies \text{generalization}.$$

Interpretation

For binary classifiers, VC dimension provides a worst-case combinatorial measure of capacity, which in turn controls the Rademacher complexity and therefore yields generalization bounds.

Why go beyond VC?

VC dimension is particularly natural for binary classification: it measures the ability of a class to realize different *labeling patterns*. For real-valued predictors, however, the picture is more subtle. There are natural extensions of VC theory, such as **pseudo-dimension** and **fat-shattering dimension**, but these remain essentially dimension-based measures of capacity.

A revealing example

For linear regression,

$$f_{\beta}(\mathbf{x}) = \mathbf{x}^{\top} \beta, \quad \mathbf{x} = (1, \mathbf{x}), \quad \beta \in \mathbb{R}^{p+1},$$

the pseudo-dimension is $p + 1$. Under suitable boundedness assumptions, this leads schematically to uniform-deviation bounds of order $\sqrt{p/n}$.

Thus the classical regime $p \ll n$ is perfectly compatible with generalization, whereas a purely dimension-based bound becomes uninformative when $p \gg n$.

The modern question

Can complexity be controlled by the **geometry** of the predictor class rather than merely by its parameter dimension?

Linear predictors: geometry instead of dimension

Consider the class of linear predictors

$$\mathcal{F}_B = \{\mathbf{x} \mapsto \langle \mathbf{w}, \mathbf{x} \rangle : \|\mathbf{w}\|_2 \leq B\}.$$

For a fixed sample $\mathbf{X}_1, \dots, \mathbf{X}_n$,

$$\begin{aligned}\hat{\mathfrak{R}}_n(\mathcal{F}_B) &= \frac{1}{n} \mathbb{E}_\epsilon \left[\sup_{\|\mathbf{w}\|_2 \leq B} \left| \left\langle \mathbf{w}, \sum_{i=1}^n \epsilon_i \mathbf{X}_i \right\rangle \right| \middle| \mathbf{X}_1, \dots, \mathbf{X}_n \right] = \frac{B}{n} \mathbb{E}_\epsilon \left[\left\| \sum_{i=1}^n \epsilon_i \mathbf{X}_i \right\|_2 \middle| \mathbf{X}_1, \dots, \mathbf{X}_n \right] \\ &\leq \frac{B}{n} \left(\sum_{i=1}^n \|\mathbf{X}_i\|_2^2 \right)^{1/2}.\end{aligned}$$

If, in addition, $\|\mathbf{X}_i\|_2 \leq R$, $i = 1, \dots, n$, then

$$\boxed{\hat{\mathfrak{R}}_n(\mathcal{F}_B) \leq \frac{BR}{\sqrt{n}}}.$$

Interpretation

The bound depends on the geometry of the class (through the norm constraint B) and on the input norm bound R , but not explicitly on the number p of covariates.

From logistic regression to deep networks

In the previous lecture, logistic regression

$$P(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \sigma(\beta_0 + \beta_{\bullet}^{\top} \mathbf{x}), \quad \beta_{\bullet} = (\beta_1, \dots, \beta_p), \quad \sigma = \text{logit}^{-1},$$

was extended by replacing its linear predictor with a one-hidden-layer neural network. More generally, let

$$d_0 = p, \quad d_1, \dots, d_{L-1}, \quad d_L = 1,$$

where d_{ℓ} is the number of units in layer ℓ . Starting from $\mathbf{h}_0 = \mathbf{x} \in \mathbb{R}^{d_0}$, define recursively

$$\mathbf{h}_{\ell} = \phi(W_{\ell} \mathbf{h}_{\ell-1} + \mathbf{b}_{\ell}), \quad W_{\ell} \in \mathbb{R}^{d_{\ell} \times d_{\ell-1}}, \quad \ell = 1, \dots, L-1, \quad \phi = \text{activation}.$$

The output layer produces the scalar score

$$f_{W,b}(\mathbf{x}) = W_L \mathbf{h}_{L-1} + b_L = W_L \phi(W_{L-1} \phi(\dots \phi(W_1 \mathbf{x} + \mathbf{b}_1) \dots) + \mathbf{b}_{L-1}) + b_L,$$

and, for binary classification, we retain the output sigmoid $P(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \sigma(f_{W,b}(\mathbf{x}))$.

Parameter count

Each layer contains $d_{\ell} d_{\ell-1}$ weights and d_{ℓ} biases. Hence

$$N_{\text{par}} = \sum_{\ell=1}^L d_{\ell}(d_{\ell-1} + 1) \xrightarrow{L=2, d_0=p, d_1=m, d_2=1} N_{\text{par}} = m(p + 2) + 1.$$

Neural networks: complexity through composition

To isolate the mechanism controlling complexity, omit the biases for simplicity and write the network as

$$f_{\mathbf{w}} = W_L \circ \phi \circ W_{L-1} \circ \cdots \circ \phi \circ W_1.$$

Assume that the activation ϕ is 1-Lipschitz, with $\phi(0) = 0$, and that $\|W_\ell\|_{\text{op}} \leq S_\ell$, $\ell = 1, \dots, L$.

Two mechanisms propagate complexity through the network:

Layerwise control

- **Linear layers:** multiplication by W_ℓ can amplify fluctuations by at most its operator norm S_ℓ .
- **Activation layers:** since ϕ is Lipschitz, the **contraction principle** prevents the nonlinearity from creating an uncontrolled increase in Rademacher complexity.

Peeling the network one layer at a time therefore contributes an operator-norm factor S_ℓ , together with additional layerwise geometric terms. Schematically,

$$\hat{\mathfrak{R}}_n(\mathcal{F}_{\text{NN}}) \lesssim \left(\prod_{\ell=1}^L S_\ell \right) \times \{\text{remaining geometric factors}\}.$$

The key point

Neural-network complexity is governed by how the successive layers **amplify fluctuations**, not merely by N_{par} .

Norm-based complexity of neural networks

For the class of depth- L networks above, omit the biases for simplicity and assume that: i) ϕ is 1-Lipschitz and positively homogeneous, as for ReLU; ii) $\|W_\ell\|_{\text{Frob}} \leq F_\ell$, $\ell = 1, \dots, L$; iii) $\|\mathbf{X}_i\|_2 \leq R$, $i = 1, \dots, n$.

A representative Rademacher bound (Golowich–Rakhlin–Shamir, 2018)

For the corresponding class $\mathcal{F}_{\text{NN}}$,

$$\hat{\mathfrak{R}}_n(\mathcal{F}_{\text{NN}}) \leq 4R \sqrt{\frac{L}{n}} \prod_{\ell=1}^L F_\ell.$$

Several features are worth noting:

- the complexity decays at the usual $n^{-1/2}$ rate; for **fixed** depth and norm bounds, it therefore vanishes as $n \rightarrow \infty$, yielding uniform convergence and hence generalization;
- there is no explicit dependence on the layer widths $d_1, \dots, d_{L-1}$, nor on N_{par} ;
- complexity is instead controlled by the norms of the learned weight matrices and by their propagation through depth.

Effective capacity

A network may contain a huge number of parameters and still belong to a class of controlled statistical complexity if its weights remain sufficiently constrained.

A missing piece: optimization

So far we have treated empirical risk minimization as if $\hat{f}_n = \arg \min_{f \in \mathcal{F}} \hat{R}_n(f)$ were actually available. For neural networks, however, the predictor is parametrized by $\theta = (\mathbf{W}, \mathbf{b})$, and exact empirical risk minimization would require finding an exact minimizer $\hat{\theta}_n$ of $\theta \mapsto \hat{R}_n(\theta) = n^{-1} \sum_{i=1}^n \ell(f_\theta(\mathbf{X}_i), \mathbf{Y}_i)$. This is typically a high-dimensional, nonconvex optimization problem. In practice, gradient-based algorithms produce some $\tilde{\theta}_n$, rather than an exact empirical minimizer.

Optimization error

If $\varepsilon_{\text{opt}} = \hat{R}_n(\tilde{\theta}_n) - \inf_{\theta} \hat{R}_n(\theta)$, then

$$R(f_{\tilde{\theta}_n}) - R(f_{\mathcal{F}}^*) \leq \varepsilon_{\text{opt}} + \sup_{f \in \mathcal{F}} |R(f) - \hat{R}_n(f)| + o_P(1).$$

Thus optimization error enters as an additional contribution to the excess population risk.

An important distinction

Complexity bounds explain why a well-trained predictor may generalize; they do not explain why gradient-based training finds such a predictor.

What have we learned about generalization?

The central statistical problem is to understand when empirical risk minimization produces a predictor whose population risk is close to optimal. The main chain of ideas developed in this lecture is

ERM $\implies$ uniform convergence $\implies$ Rademacher complexity $\implies$ capacity control $\implies$ generalization.

Three lessons emerge:

- **Uniformity has a price:** the complexity of the hypothesis class determines whether empirical averages approximate population quantities uniformly.
- **Richness creates a trade-off:** more expressive classes reduce approximation error, but make estimation and generalization harder.
- **Complexity is not merely parameter counting:** VC dimension is natural for binary classifiers, while for high-dimensional and deep models geometric quantities such as norm constraints may provide much sharper control.

Take-home message

Generalization is possible when the complexity of the predictor class grows slowly enough relative to the amount of data.

Breiman's "two cultures" now appears remarkably prescient: modern machine learning, and deep neural networks in particular, has confirmed the growing primacy of

prediction over explanation.

A successful predictor need not provide a transparent model of the data-generating mechanism. Once this shift is accepted, the fundamental statistical question becomes:

Why should prediction work on unseen data?

The theory developed in this lecture is a first step toward answering this question for modern predictive models, including neural networks: statistical learning provides tools for understanding when predictive performance can generalize beyond the training sample.

But generalization is only part of the story. In the next lecture, we ask how the uncertainty associated with an individual prediction can be quantified in a statistically meaningful way.

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From classical statistics to statistical learning: a probabilistic overview

IV — Conformal Prediction: Exchangeability and Finite-Sample Guarantees

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From independence to exchangeability

Throughout the previous lectures, the statistical paradigm was based on an i.i.d. sample

$$\mathbf{Z}_1, \dots, \mathbf{Z}_n \stackrel{\text{iid}}{\sim} P, \quad \mathbf{Z}_i = (\mathbf{X}_i, Y_i).$$

For Conformal Prediction, however, independence is stronger than what is actually needed.

Exchangeability

A finite sequence of random variables $W_1, \dots, W_m$ is called *exchangeable* if, for every permutation π of $\{1, \dots, m\}$,

$$(W_1, \dots, W_m) \stackrel{d}{=} (W_{\pi(1)}, \dots, W_{\pi(m)}).$$

Thus,

$$\text{i.i.d.} \implies \text{exchangeable},$$

but the converse is false.

Key idea

Exchangeability replaces independence by a symmetry principle: the joint distribution does not depend on the ordering of the observations.

Exchangeability beyond independence

Exchangeability allows dependence among the observations. A particularly transparent example is provided by binary random variables: $W_1, W_2, \dots \in \{0, 1\}$.

Suppose there exists a random variable $\Theta \in [0, 1]$ such that, conditionally on $\Theta = \theta$,

$$W_1, W_2, \dots \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta).$$

Then, for every $m \geq 1$ and $w_1, \dots, w_m \in \{0, 1\}$,

$$\mathbb{P}(W_1 = w_1, \dots, W_m = w_m) = \int_0^1 \theta^{\sum_{i=1}^m w_i} (1 - \theta)^{m - \sum_{i=1}^m w_i} d\mu(\theta),$$

where μ is the distribution of Θ . Since the right-hand side depends on $w_1, \dots, w_m$ only through $\sum_{i=1}^m w_i$, the sequence is exchangeable. Remarkably, the converse also holds.

de Finetti's theorem

Every infinite exchangeable binary sequence admits such a mixture-of-Bernoulli representation.

Beyond independence

Exchangeability may involve substantial dependence: the variables become independent only after conditioning on the latent parameter Θ .

Let $R_1, \dots, R_{m+1}$ be exchangeable real-valued random variables, and assume for simplicity that ties occur with probability zero.

For each j , define the rank of R_j among $R_1, \dots, R_{m+1}$ by

$$\text{rank}(R_j) = 1 + \sum_{i \neq j} 1\{R_i < R_j\}.$$

By exchangeability, no index is probabilistically distinguished from the others. Hence

$$\mathbb{P}(\text{rank}(R_{m+1}) = k) = \frac{1}{m+1}, \quad k = 1, \dots, m+1 \implies \text{rank}(R_{m+1}) \sim \text{Unif}\{1, \dots, m+1\}.$$

The rank principle

For exchangeable observations, the position occupied by any prescribed observation in the ordered sample is uniformly distributed.

This simple symmetry argument is the probabilistic core of Conformal Prediction.

From statistical samples to prediction

We now return to the general supervised learning setting. Let

$$\mathbf{Z}_i = (\mathbf{X}_i, Y_i), \quad i = 1, \dots, n,$$

be an exchangeable sample from a joint distribution on $\mathcal{X} \times \mathcal{Y}$.

Our goal is no longer only to estimate features of the underlying distribution, but to predict the response Y associated with a new covariate value $\mathbf{X}$.

In split Conformal Prediction, the available sample is partitioned into two parts:

$$\underbrace{\mathbf{Z}_1, \dots, \mathbf{Z}_{n_{\text{tr}}}}_{\text{training sample}} \quad \text{and} \quad \underbrace{\mathbf{Z}_{n_{\text{tr}}+1}, \dots, \mathbf{Z}_n}_{\text{calibration sample}}.$$

The training sample is used to construct a predictor

$$\hat{f} : \mathcal{X} \rightarrow \mathcal{Y},$$

while the calibration sample will be used to quantify the uncertainty of its predictions.

Main question

Given a new covariate $\mathbf{X}_{n+1}$, can we construct a set $C(\mathbf{X}_{n+1})$ such that

$$\mathbb{P}(Y_{n+1} \in C(\mathbf{X}_{n+1})) \geq 1 - \alpha?$$

From predictions to nonconformity scores

Fix the predictor $\hat{f}$ obtained from the training sample.

To measure how well a candidate pair $(\mathbf{x}, y)$ conforms to the fitted predictor, introduce a *nonconformity score*

$$S(\mathbf{x}, y; \hat{f}) \in \mathbb{R},$$

with larger values indicating poorer agreement with the prediction. For instance, for regression the simplest choice is the absolute residual

$$S(\mathbf{x}, y; \hat{f}) = |y - \hat{f}(\mathbf{x})|.$$

Thus, for the m observations in the calibration sample, define

$$R_i = S(\mathbf{X}_i, Y_i; \hat{f}), \quad i = 1, \dots, m.$$

For a new observation $(\mathbf{X}_{m+1}, Y_{m+1})$, define similarly

$$R_{m+1} = S(\mathbf{X}_{m+1}, Y_{m+1}; \hat{f}).$$

Key observation

Conditionally on the training sample, the scores $R_1, \dots, R_m, R_{m+1}$ are exchangeable.

From ranks to empirical quantiles

Let $R_1, \dots, R_m$ be the calibration scores, and denote their order statistics by

$$R_{(1)} \leq R_{(2)} \leq \dots \leq R_{(m)}.$$

Consider now an additional score R_{m+1} , exchangeable with the calibration scores. Assuming again that there are no ties,

$$R_{m+1} \leq R_{(k)} \iff \text{rank}(R_{m+1}) \leq k, \quad k = 1, \dots, m.$$

By the uniform rank principle,

$$\mathbb{P}(R_{m+1} \leq R_{(k)}) = \frac{k}{m+1}.$$

Thus, given a target miscoverage level $\alpha \in (0, 1)$, choose

$$k = \lceil (m+1)(1-\alpha) \rceil,$$

provided $k \leq m$. Then

$$\mathbb{P}(R_{m+1} \leq R_{(k)}) = \frac{k}{m+1} \geq 1 - \alpha.$$

Conformal quantile

The threshold $R_{(k)}$, with $k = \lceil (m+1)(1-\alpha) \rceil$, is the finite-sample correction of the usual empirical $(1-\alpha)$ -quantile.

Recall that the calibration scores are

$$R_i = S(\mathbf{X}_i, Y_i; \hat{f}), \quad i = 1, \dots, m,$$

and let

$$\hat{q}_{1-\alpha} = R_{(k)}, \quad k = \lceil (m+1)(1-\alpha) \rceil.$$

For a new covariate value $\mathbf{x}$, define the prediction set

$$C(\mathbf{x}) = \left\{ y \in \mathcal{Y} : S(\mathbf{x}, y; \hat{f}) \leq \hat{q}_{1-\alpha} \right\}.$$

Thus,

$$Y_{m+1} \in C(\mathbf{X}_{m+1}) \iff R_{m+1} \leq \hat{q}_{1-\alpha}.$$

By the rank argument,

$$\mathbb{P}(Y_{m+1} \in C(\mathbf{X}_{m+1})) \geq 1 - \alpha.$$

Finite-sample coverage

The prediction set $C(\mathbf{X}_{m+1})$ contains the new response with probability at least $1 - \alpha$, under exchangeability.

The split conformal interval

In regression, take the nonconformity score

$$S(\mathbf{x}, y; \hat{f}) = |y - \hat{f}(\mathbf{x})|.$$

Hence, the calibration scores are simply the absolute residuals

$$R_i = |Y_i - \hat{f}(\mathbf{X}_i)|, \quad i = 1, \dots, m.$$

Let

$$\hat{q}_{1-\alpha} = R_{(k)}, \quad k = \lceil (m+1)(1-\alpha) \rceil.$$

Then

$$S(\mathbf{x}, y; \hat{f}) \leq \hat{q}_{1-\alpha} \iff |y - \hat{f}(\mathbf{x})| \leq \hat{q}_{1-\alpha}.$$

Therefore, the conformal prediction set is the interval

$$C(\mathbf{x}) = [\hat{f}(\mathbf{x}) - \hat{q}_{1-\alpha}, \hat{f}(\mathbf{x}) + \hat{q}_{1-\alpha}].$$

Coverage guarantee

For a new exchangeable observation $(\mathbf{X}_{m+1}, Y_{m+1})$,

$$\mathbb{P}(Y_{m+1} \in C(\mathbf{X}_{m+1})) \geq 1 - \alpha.$$

The conformal guarantee

$$\mathbb{P}(Y_{m+1} \in C(\mathbf{X}_{m+1})) \geq 1 - \alpha$$

is a *marginal* coverage statement: it averages simultaneously over the randomness in

$$\mathbf{X}_{m+1}, \quad Y_{m+1},$$

as well as over the data used to construct the prediction set.

In general, it does *not* imply the stronger conditional guarantee

$$\mathbb{P}(Y_{m+1} \in C(\mathbf{X}_{m+1}) \mid \mathbf{X}_{m+1} = \mathbf{x}) \geq 1 - \alpha$$

for every $\mathbf{x} \in \mathcal{X}$.

An important distinction

Conformal Prediction provides exact finite-sample marginal validity under exchangeability, but exact distribution-free conditional coverage is, in general, unattainable without additional assumptions.

Conformal versus classical prediction intervals

Recall the classical linear regression model discussed in Lecture II (with the intercept omitted):

$$\mathbf{Y} = \mathbf{X}^\top \beta + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

Under the usual assumptions, a prediction interval for a new response at $\mathbf{X} = \mathbf{x}$ has the form

$$\mathbf{C}_{\text{class}}(\mathbf{x}) = \left[\hat{f}(\mathbf{x}) \mp t_{n-p, 1-\alpha/2} \hat{\sigma} \sqrt{1 + \mathbf{x}^\top \mathbf{S} \mathbf{x}} \right], \quad \mathbf{S} = (\mathcal{X}^\top \mathcal{X})^{-1},$$

and its coverage guarantee is conditional on the covariate value:

$$\mathbb{P}(Y_{\text{new}} \in \mathbf{C}_{\text{class}}(\mathbf{x}) \mid \mathbf{X}_{\text{new}} = \mathbf{x}) = 1 - \alpha,$$

provided the linear-Gaussian model is correctly specified.

By contrast, under exchangeability split Conformal Prediction requires no assumptions of linearity, Gaussianity, or homoskedasticity, but guarantees instead

$$\mathbb{P}(Y_{\text{new}} \in \mathbf{C}_{\text{CP}}(\mathbf{X}_{\text{new}})) \geq 1 - \alpha.$$

The trade-off

Classical prediction intervals obtain stronger conditional guarantees from stronger model assumptions; Conformal Prediction obtains distribution-free finite-sample validity from the weaker assumption of exchangeability, at the level of marginal coverage.

The conformal coverage guarantee is remarkably robust: it does not require the predictor $\hat{f}$ to be well specified, or even particularly accurate. If the prediction rule is poor, the calibration scores $R_i = |Y_i - \hat{f}(\mathbf{X}_i)|$ will tend to be large, and so will the conformal quantile $\hat{q}_{1-\alpha}$.

Thus, validity is preserved, but the prediction interval

$$C(\mathbf{x}) = [\hat{f}(\mathbf{x}) - \hat{q}_{1-\alpha}, \hat{f}(\mathbf{x}) + \hat{q}_{1-\alpha}]$$

may become very wide.

Two separate questions

Validity: $\mathbb{P}(Y_{\text{new}} \in C(\mathbf{X}_{\text{new}})) \geq 1 - \alpha,$

Efficiency: How small can $C(\mathbf{x})$ be while preserving validity?

Uniform prediction band in the split case

In the simplest split conformal construction, the interval width $2\hat{q}_{1-\alpha}$ does not depend on $\mathbf{x}$.

The basic split conformal interval has constant width because it uses the absolute residual

$$R_i = |Y_i - \hat{f}(\mathbf{X}_i)|.$$

When the conditional variability of Y changes with $\mathbf{x}$, a more adaptive choice is to normalize the residuals by an estimate of the local scale:

$$R_i = \frac{|Y_i - \hat{f}(\mathbf{X}_i)|}{\hat{\sigma}(\mathbf{X}_i)}.$$

If $\hat{q}_{1-\alpha}$ is the corresponding conformal quantile of the normalized calibration scores then the prediction interval becomes

$$C(\mathbf{x}) = \left[\hat{f}(\mathbf{x}) - \hat{q}_{1-\alpha} \hat{\sigma}(\mathbf{x}), \hat{f}(\mathbf{x}) + \hat{q}_{1-\alpha} \hat{\sigma}(\mathbf{x}) \right].$$

Local adaptation yields a variable width band

The interval width

$$2\hat{q}_{1-\alpha}\hat{\sigma}(\mathbf{x})$$

now depends on $\mathbf{x}$: it is wider in regions where prediction is more difficult and narrower where the response is more stable.

Beyond regression: conformal classification

Suppose now that $Y \in \{1, \dots, K\}$, and that the fitted classifier provides estimated class probabilities

$$\hat{p}_y(\mathbf{x}) = \hat{\mathbb{P}}(Y = y \mid \mathbf{X} = \mathbf{x}).$$

A natural nonconformity score is

$$S(\mathbf{x}, y; \hat{f}) = 1 - \hat{p}_y(\mathbf{x}),$$

so that unlikely labels receive larger scores.

For the calibration observations, define

$$R_i = 1 - \hat{p}_{Y_i}(\mathbf{X}_i), \quad i = 1, \dots, m,$$

and let $\hat{q}_{1-\alpha}$ be the corresponding conformal quantile.

Then the conformal prediction set is

$$C(\mathbf{x}) = \{y \in \{1, \dots, K\} : 1 - \hat{p}_y(\mathbf{x}) \leq \hat{q}_{1-\alpha}\}.$$

Equivalently,

$$C(\mathbf{x}) = \{y : \hat{p}_y(\mathbf{x}) \geq 1 - \hat{q}_{1-\alpha}\}.$$

Coverage guarantee

Under exchangeability,

$$\mathbb{P}(Y_{\text{new}} \in C(\mathbf{X}_{\text{new}})) \geq 1 - \alpha.$$

The finite-sample validity of Conformal Prediction rests on a remarkably weak but essential assumption:

$$(\mathbf{Z}_1, \dots, \mathbf{Z}_n, \mathbf{Z}_{n+1}), \quad \mathbf{Z}_i = (\mathbf{X}_i, Y_i), \quad \text{are exchangeable.}$$

This assumption may fail, for instance, under

temporal dependence, distribution shift, or covariate shift.

In such settings, the usual conformal guarantee

$$\mathbb{P}(Y_{n+1} \in C(\mathbf{X}_{n+1})) \geq 1 - \alpha$$

need not hold without further adjustments.

Moreover, even under exchangeability, standard Conformal Prediction provides *marginal*, not pointwise conditional, coverage.

What Conformal Prediction does, and does not, guarantee

CP provides distribution-free, finite-sample predictive validity under exchangeability. It does not automatically guarantee conditional coverage, narrow prediction sets, or robustness to distribution shift.

Take-home message

Conformal Prediction rests on a simple but powerful probabilistic principle:

$$\text{exchangeability} \implies \text{uniform ranks.}$$

This leads directly to finite-sample predictive guarantees through empirical quantiles of conformity scores:

$$\mathbb{P}(Y_{\text{new}} \in C(\mathbf{X}_{\text{new}})) \geq 1 - \alpha.$$

The method is remarkably general:

any predictor + a suitable conformity score + calibration

can be turned into a valid predictive procedure under exchangeability.

At the same time, one should keep separate three distinct issues:

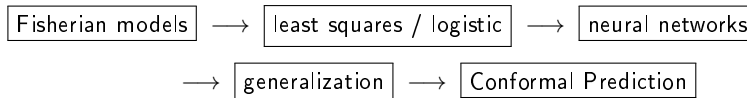
validity	efficiency	conditional coverage
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Main lesson

Conformal Prediction converts a symmetry assumption on the data into distribution-free, finite-sample uncertainty quantification.

Final perspective: from statistical inference to predictive learning

The lectures have traced a gradual shift in statistical methodology:



Classical statistical inference relies heavily on asymptotic fluctuations, $\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, \Sigma)$, which provide standard errors, confidence intervals and hypothesis tests, whereas **modern learning theory** places greater emphasis on a different question, $\hat{R}_n(f) \approx R(f)$, controlled through uniform laws of large numbers and complexity of the function class. This trajectory echoes Breiman's distinction between the **data modeling culture** and the **algorithmic modeling culture**, with modern machine learning placing increasing emphasis on **prediction** rather than **parameter interpretation**.

A change of emphasis

Prediction and generalization extend more naturally to highly complex models, whereas classical parameter-based inference does not extend as directly: neural networks are typically non-identifiable and may define singular statistical models, so the usual CLT-based asymptotic theory cannot simply be carried over. Conformal Prediction completes this shift: even for a complex black-box predictor, predictive uncertainty can be quantified without recovering classical parametric inference.

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